Big Data Analytics Final Project Report

Introduction

Within the United States of America, healthcare quality within hospitals is always under constant scrutiny due to the delicate challenge of balancing patient care with cost efficiency. Among the many metrics used to evaluate hospitals, one of the most critical measures is the patient readmission rate, representing how often a patient needs to be readmitted for the same health-related issue. The need for frequent readmissions may highlight gaps in the overall healthcare system in place at the hospital, relating to critical issues such as lack of strategic planning, misdiagnosis, and ineffective treatment.

Hospital quality assessments are particularly relevant for four common and costly conditions: pneumonia, heart attack, knee replacement, and heart failure. These four conditions often involve high levels of readmission rates, creating a significant impact on patient well-being and overall healthcare costs. When determining which hospital provides the most effective care, it is essential to look at trends in both readmission rates as well as treatment fees incurred by patients, specifically for all four of these common illnesses. Such analysis becomes crucial in making the next steps for improvements from both a business and healthcare perspective, helping to determine which hospitals need the most assistance and government interventions.

This report highlights various analytical approaches to evaluate hospitals across the United States to better understand readmission rates. The objective is to determine which factors create the most significant impact on readmission rates and which hospitals are in the most need of aid to lower the overall rate across the country.

Variables

Two datasets from the United States' Centres for Medicare and Medicaid Services website were used to evaluate the question above: Hospital Readmissions Reduction Program, and Hospital Payment and Value of Care. Several numerical variables were further evaluated. Firstly, the expected rate, which represented the benchmark 30-day readmission rate for a hospital with a similar patient population. Secondly, the Predicted Rate, which adjusted for hospital-specific factors. Thirdly, the expected readmission ratio, which was calculated as predicted rate divided by expected rate, determined whether a hospital exceeds or falls below the expected standard. Fourthly, payment referred to the average payment received for services, with lower estimates and higher estimates defining the confidence interval for estimated payments. These variables provided a comprehensive understanding of hospital quality, financial metrics, and patient outcomes (Table 1).

The classification model aimed to assess hospital quality based on the Excess Readmission Ratio (ERR), which compared a hospital's predicted and expected 30-day readmission rates. The target variable, Quality, was binary, where 1 (Bad Quality) represented hospitals with higher-than-expected readmission rates (ERR $\geq$ 1), while 0 (Good Quality) indicated hospitals with lower-than-expected readmission rates (ERR $<$ 1) (Table 1).

Several categorical variables provided contextual information about the hospitals. State represented the geographical location, which may influence patient demographics and healthcare access. Mortality categorized hospitals based on mortality performance compared to the national average (Worse, Average, Better). Similarly, Expenditure classified hospitals by spending levels (Higher, Average, Lower), reflecting resource allocation and cost efficiency. Measure_Category specified the conditions for which readmission is being evaluated, including pneumonia, heart attack, hip/knee replacement, and heart failure. These categories helped to differentiate hospitals based on patient case types (Table 1).

Exploratory Data Analysis

In Azure Databricks, the pandas library was imported to merge and prepare both datasets for visualization. Each medical condition's data frame was divided into separate data frames and then concatenated into a complete pandas data frame. Then, using Matplotlib and Seaborn, different visualizations regarding trends in the combined data frame were made.

The excess readmission ratio was then modelled by a normal distribution. There was a mean of 1.0 with a standard deviation of 0.1. It became apparent that the range of ERR across all hospitals in the United States was close, but in a practical sense, the difference mattered. For instance, assuming each hospital had the same number of patients, 100 patients, that had similar medical conditions, a hospital within the 1st standard deviation would see either 10 additional patients or fewer patients. Hospitals with high ERRs may face financial penalties, while those with low ERRs may serve as benchmarks for best practices (Figure 1).

The different conditions and their payment costs were then analyzed. Examining Heart Attack, Pneumonia, Hip/Knee Replacement, and Heart Failure, it was apparent that the average cost of payment progressively increased from left to right, with a significant increase for hip and knee replacement. This variation was likely due to differences in treatment complexity, surgical interventions, and recovery time. Hip/Knee Replacement involves extended hospital stays, physical therapy, and follow-up procedures, which naturally elevate costs, and therefore have higher payment values than heart attacks and pneumonia. Physical therapy and follow-up procedures would require the patient to visit the hospital repeatedly, which may lead to readmissions. Additionally, heart failure often requires intensive care, emergency interventions, and ongoing medication management. The severity and extended cost of surgery may result in some patients opting out of complete treatment, either due to financial barriers or concerns over surgical risk. Thus, the payment for very serious conditions like pneumonia and heart attacks has, on average, a lower cost than a hip/knee replacement (Figure 2).

Next, comparing what is deemed a good hospital (ERR < 1) and a bad hospital (ERR >= 1), there was a very slim difference between the payments in each quartile. Bad hospitals tend to have a higher payment, but with it being so marginal, it was difficult to conclude with certainty that bad hospitals charged more than good hospitals for each condition (Figure 3).

Finally, a violin plot showing the distribution of the excess readmission ratio across the four different conditions was made. Notably, Hip/Knee Replacement exhibited the widest distribution, indicating greater variability in hospital performance for surgical readmissions. In contrast, Pneumonia and Heart Attacks showed tighter, more concentrated distributions, suggesting more consistency in hospital readmission outcomes. The medians for all conditions hover around an ERR of 1.0. Overall, readmission performance varied not only between hospitals but also across different types of care (Figure 4).

Classification

Modelling Choices

This project implemented seven classification models to predict hospital quality based on the Excess Readmission Ratio (ERR) and associated features. The models include Gradient Boosted Trees (GBT), Decision Tree, Logistic Regression, One-vs-Rest (OvR) with Logistic Regression, Linear Support Vector Classifier (Linear SVC), Random Forest, and Naïve Bayes.

Feature Processing and Scaling Strategies

Feature engineering was critical in ensuring that categorical and numerical variables were handled correctly. The categorical features (State, Mortality, Expenditure, Measure_Category) were transformed using StringIndexer and One-Hot Encoding to convert them into a numerical format interpretable by machine learning algorithms. The numerical features (Payment, Lower Estimate, Higher Estimate) remained unchanged, except for models that required scaling. A VectorAssembler was used to combine all preprocessed features into a single feature vector before model training. A pipeline was adopted to facilitate the process and improve efficiency.

A key distinction in preprocessing was that Logistic Regression, One-vs-Rest (OvR), and Linear SVC were trained on scaled features using a scaler. In contrast, the remaining models (GBT, Decision Tree, Random Forest, and Naïve Bayes) used the original data with only encoding and feature indexing applied. The rationale behind this decision lies in the sensitivity of certain models to feature magnitudes.

Cross-Validation & Hyperparameter Tuning

Rather than arbitrarily selecting hyperparameters, a grid search was used with cross-validation to identify the optimal settings. A pipeline was also adopted to facilitate the process and improve efficiency. K-fold cross-validation was used (k=3) to evaluate various hyperparameter combinations systematically. The data was split into 80% for training and 20% for testing.

To identify the optimal hyperparameters, a paramGrid search was employed, which exhaustively tests a specified range of hyperparameters. For each combination, the model was trained and evaluated using cross-validation.

The implementation of this process via Spark ML's CrossValidator function automated the grid search. The function first evaluated the model on each fold and then identifies the best-performing model and its corresponding hyperparameters. The combination that yielded the highest AUC score was selected as the best configuration.

The Best Model Selection

After performing grid search-based cross-validation, the optimal hyperparameters for each model were modified. The model that achieved the highest AUC score on the test set was selected as the final model out of seven options. The Random Forest model had the highest AUC (0.632), slightly outperforming Logistic Regression (0.630). However, these values indicated that the models have only a moderate ability to distinguish between high- and low-quality hospitals. The Decision Tree identifies most of the bad hospitals (high TP = 694) but also mislabels many good hospitals as bad (FP = 627), leading to potential unnecessary interventions. Logistic Regression and Random Forest have a more balanced classification but may miss some bad hospitals. If avoiding wrongly flagging good hospitals is the priority, Random Forest or Logistic Regression would be a good fit, while if catching as many bad hospitals as possible is the goal, Decision Tree appeared to be the best tool.

Clustering

Clustering was also used to uncover insightful observations to answer the key research question. Specifically, it was used to explore the relationship between payments and excess readmission rates. For example, if a hospital required high payment for the duration of the treatment and had low readmission rates, this might indicate that although the hospital's treatments were expensive, they were also effective.

However, it is also possible for high readmission rates to occur because patients are discharged early due to an inability to pay for an extensive and costly treatment. Unresolved medical issues are more likely to result in higher readmission rates. Additionally, if patients cannot afford follow-up care, such as rehabilitation and home care services, to fully resolve their medical issues, they may be more likely to experience higher readmission rates.

Clustering was first performed by joining the readmissions and payments datasets. The original dataset had column names with spaces in them, which required underscores to be added to replace the spaces. For example, the column name "Facility ID" needed to be changed to "Facility_ID" for the datasets to successfully join into one table and run on Databricks. The data was preprocessed by converting date columns into numeric columns, encoding categorical variables, filling missing numeric values with 0 and categorical values with -1, combining features into a vector, and standardizing features. The silhouette score was calculated to determine the optimal number of clusters. Figure 5 shows that the silhouette score is highest for four clusters; therefore, that is the optimal number of clusters to be used. The model was then trained based on the optimal silhouette score, and the standardized dataset was used to fit the model. Predictions using the model and cluster centres were then calculated. A scatterplot was created to visualize clusters, with the x-axis being readmission ratios and the y-axis being the payment amount. A colour bar was also created to represent different clusters with different colours. Finally, the scatterplot was interpreted by calculating various average values for each cluster, including the average excess readmission ratio and average payments.

Further analysis of each cluster was performed to gain a more comprehensive understanding of the model. Investigations were conducted on the types of diseases that were treated and where the hospital was located for each cluster. For example, if a hospital typically treats chronic illnesses requiring expensive medication or medical procedures, it would give more context to understand why that hospital experiences high readmission rates and payments. Additionally, if a hospital is located in a state that does not provide sufficient health benefits, it can be implied that hospitals with high readmission rates and high payments do not necessarily overcharge their patients. Since disease type and state were categorical variables, the mode was evaluated to find a common theme for each cluster.

Figure 6 depicts a scatterplot showing four clusters of hospitals evaluated on the relationship between the excess readmission ratio and payment levels. The four clusters appear to be close to each other, which implies that they have similar average payment and average excess readmission ratios, and they are represented in Table 2.

There are slight differences between clusters. For example, cluster 1 has the highest average excess readmission ratio and average payment. From a business perspective, this could mean that the hospital is ineffective at treating patients while still charging them high prices, suggesting that treatments are overpriced. However, high readmission rates and high payment can also occur when a majority of patients in the hospital suffer from chronic conditions (e.g. heart failure), which require expensive treatment (e.g., prescriptions, medical procedures) during each visit. Therefore, it is recommended that the hospital conduct further investigations on patient profiles to understand why they are experiencing high readmission rates.

Cluster 3 has the lowest average excess readmission ratio and the lowest average payment. From a business perspective, this could mean that the hospital is highly effective at treating patients while charging them low prices. For example, the hospital could be focusing on preventative care, helping patients identify illnesses before requiring expensive hospitalization. This is relatively cost-efficient compared to large medical procedures and prescriptions used in later-stage illnesses. Preventive care

could lead to reduced ER visits, resulting in lower readmission rates. Therefore, it is recommended that further investigation be conducted on the types of services that hospitals provide to patients as another way to understand what factors contribute to readmission.

Table 3 shows that two types of diseases are the most common across all four clusters. Clusters 1 and 3 have the same mode for payment_measure_name_indexed, which represents the disease associated with a payment made at each hospital. As mentioned, cluster 1 has the highest average excess readmission ratio and the highest average payment, while cluster 3 has the lowest average excess readmission ratio and the lowest average payment. Therefore, there is little relationship among disease, payments, and readmission rates. Table 3 also shows that clusters 1 and 4 have the same mode for state. Both clusters have almost the same average payment and very similar average readmission rates. It could be implied that there is a relationship between state and payment, suggesting that state health benefits directly impact how much patients have to pay for their treatments. If a state's health benefit includes wide coverage of services and higher percentage of health-related fees covered, patients may be incentivized to return to hospitals for more thorough check-ups, leading to high readmission rates. However, clusters 1 and 4 both have high average payments, which may not indicate that state health benefits are generous enough to lower health expenses enough for patients to return for thorough check-ups. However, it is still recommended that hospitals collect data on patients' health benefits, both public and private, to understand whether access to insurance can influence the ability to afford payments and how that may affect readmission rates.

Challenges

During the EDA process, from the ERR distribution graph, the distribution seems to be almost perfectly normal, with an ERR centred around 1. This normal distribution is likely due to the large sample size, including data from hospital records across the United States. Based on the statistical theory, an increased sample size tends to produce more normally distributed data that better represents the true population parameters. This characteristic of the dataset reveals the potential risk that the ERR values cluster within a narrow range. The distribution means that introducing biased data, such as data that is constantly above or below 1, would result in a significant skew to the distribution.

The biggest challenges for classification would be during the tuning process. One of the primary concerns was the computational expense of grid search with cross-validation, especially when dealing with multiple models that have many hyperparameters. Training and validating models through an exhaustive search of all hyperparameter combinations is highly resource-intensive, particularly for large-scale datasets. In distributed computing environments like PySpark, the process becomes even more constrained, often leading to significant delays in model optimization. While certain hyperparameters have minimal influence on performance, grid search methods evaluate every possible combination, resulting in heavy computational expenses. Randomized search is a common alternative that offers a more efficient way of sampling hyperparameters from predefined distributions, but it may not always yield the most optimal results. Given this project's relatively small range of hyperparameters, an exhaustive search using ParamGrid was feasible. This approach thoroughly searched all the combinations of hyperparameters, resulting in a more precise model tuning compared to randomized methods.

Addressing "not available" values under the payment column was also a challenge during clustering. With over 8,000 entries marked as "not available", it was difficult to decide on the imputation methods. Replacing with 0 or the mean cannot fully account for the payment variations across hospitals. These missing values may come from various sources, including data accessibility limitations. While

replacing those values with 0 is a temporary solution, a more robust hospital data collection system is recommended to reduce null entities and ensure data completeness.

Application

Our findings can help hospitals optimize their resource allocation by identifying clusters with high costs and high readmission rates, which can minimize unnecessary expenses and allow them to invest more in preventive care to reduce readmission rates. Similarly, classification helps hospitals to analyze if actual treatment needs are associated with excessive charges or not to prevent them from overpricing while not compromising on the quality of care and reducing unnecessary costs. Moreover, clusters with low payments and low readmission rates show better preventive care, and hospitals can apply the same measures in other areas to improve performance in diagnosing patients' conditions and lower costs. Hospitals with high payments but poor patient outcomes can examine their patients' profiles and the treatment provided. After analysis, they can refine their medical procedures to ensure cost-effective and appropriate patient care. This analysis can also help hospitals better plan their finances and support programs based on the correlations between state benefits and payment trends, which explain how and why financial assistance is important for a patient's return.

Recommendations

To help hospitals better manage quality and allocate resources more effectively, we recommend leveraging predictive models to identify institutions at higher risk of excessive readmission rates. This would enable earlier and more targeted interventions that improve outcomes while avoiding unnecessary treatment. Additionally, clustering techniques can be used to compare hospital performance with similar institutions, helping uncover inefficiencies and highlight best practices. Improving data infrastructure, particularly in the areas of financial and operational metrics, is also essential for generating accurate and actionable insights. Future data collection efforts should aim to include additional patient-level variables such as comorbidities, socioeconomic status, and access to follow-up care. This would support more personalized and effective analytics. Finally, we recommend using these insights to guide resource optimization, ensuring that higher-cost services are aligned with measurable improvements in patient outcomes.

Conclusion:

This project explored whether hospital readmission rates can be predicted using publicly available data and whether meaningful patterns in hospital performance can be uncovered through machine learning. By focusing on four prevalent conditions, including pneumonia, heart attack, hip/knee replacement, and heart failure, we examined how hospital-specific metrics, such as average payments, regional attributes, and readmission ratios, contribute to performance classification.

Among the seven classification models tested, Random Forest achieved the highest AUC score of 0.632, followed closely by Logistic Regression at 0.630. These results demonstrate the feasibility of using machine learning to differentiate hospital quality with moderate predictive accuracy. While not perfect, the models show potential for identifying at-risk institutions and informing quality oversight processes. Additionally, Decision Trees, while capturing more high-risk hospitals, produced a more significant number of false positives, reinforcing the importance of model selection when precision is critical.

Our clustering analysis revealed four distinct hospital segments based on patterns in readmission rates and average payments. Notably, Cluster 1 exhibited the highest values in both metrics, raising questions about

efficiency, while Cluster 3 reflected low payments and readmissions, suggesting a combination of effective care delivery and cost control. These findings provide a clearer understanding of the trade-offs hospitals face between cost and outcomes and illustrate the potential for segmentation to inform peer comparisons, funding models, and strategic planning.

The project also surfaced limitations that warrant consideration. A significant portion of financial data was missing or labelled “Not Available,” with over 8,000 records replaced by zero values. This likely introduced distortions and highlights the pressing need for more consistent and complete hospital reporting practices. Furthermore, while the dataset offered a strong foundation, it lacked key clinical and contextual variables, such as patient comorbidities, hospital size, staffing ratios, and community-level socioeconomic factors, that may further enhance model accuracy.

Despite these limitations, the study underscores the growing value of data science in healthcare operations. The ability to uncover patterns at scale and support evidence-based decision-making positions machine learning as a valuable tool for driving system-wide improvements. As data quality continues to improve and modelling techniques evolve, these tools could play a critical role in supporting hospital accountability, improving transparency, and shaping more equitable healthcare delivery across the system.

Figures and Tables

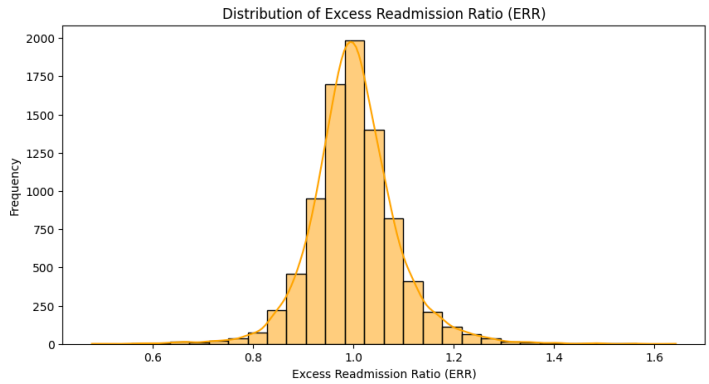

Figure 1. Distribution of Excess Readmission Ratio

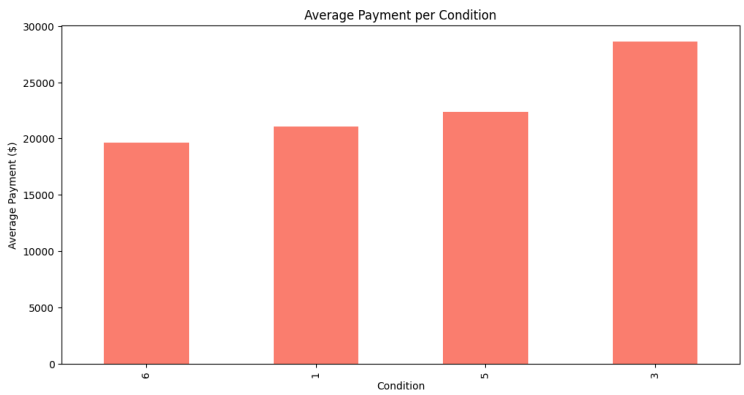

Figure 2. Average Payment per Condition. 1 = Pneumonia (p), 3 = Heart Failure (Hf), 5 = Hip/Knee Replacement (h/kr), 6 = Heart Attack (Ha)

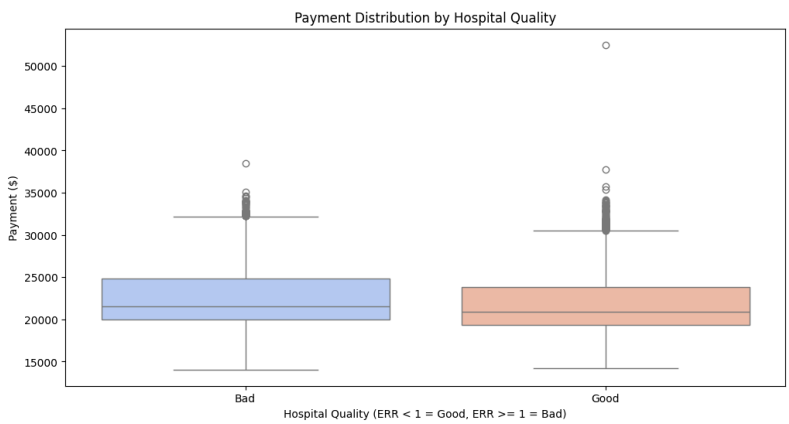

Figure 3. Payment Distribution by Hospital Quality

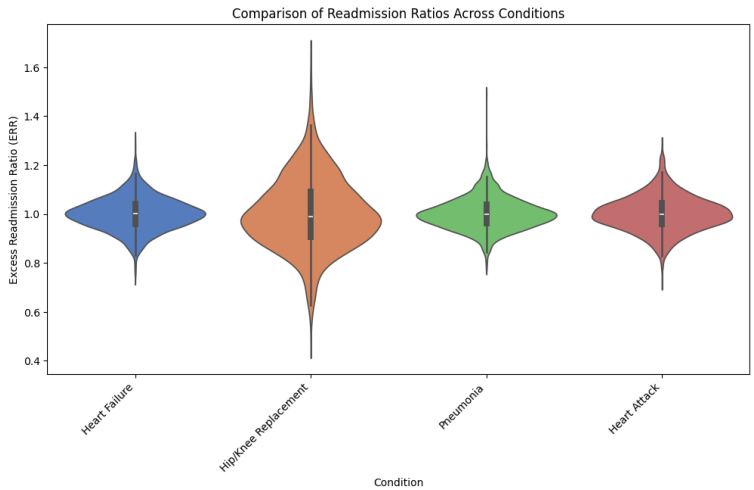

Figure 4. Comparison of Readmission Ratios Across Conditions

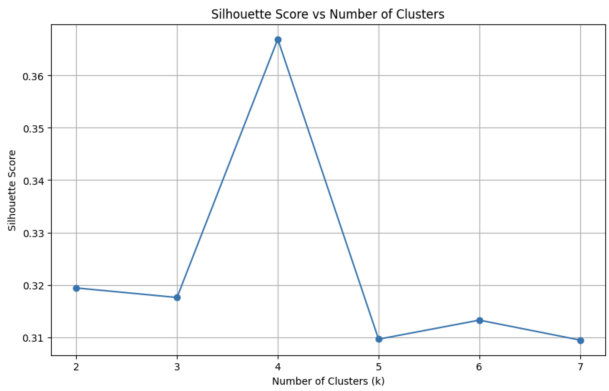

Figure 5. Silhouette Score For Each Cluster Amount

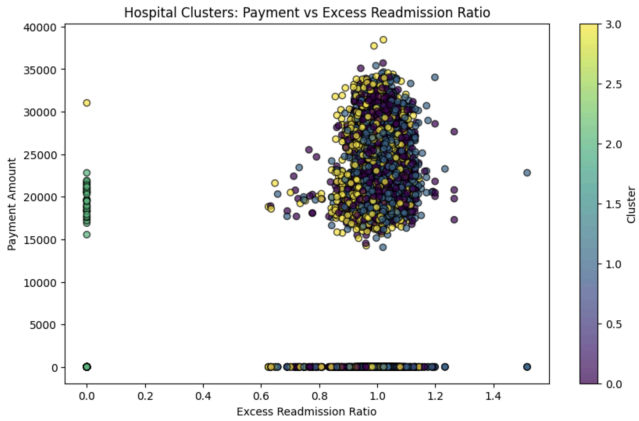

Figure 6. Cluster of Hospital Payment and Excess Readmission Ratio

Table 1. Variable types, names, and descriptions

Type	Variable	Description
Binary (Target)	Quality	1 = Bad Quality (ERR ≥ 1), 0 = Good Quality (ERR < 1)
Categorical	State	Geographical location of the hospital
	Mortality	Hospital mortality performance than the national average (Worse, Average, Better)
	Expenditure	Hospital spending levels than national average (Higher, Average, Lower)
	Measure_Category	Measured based on the type of admitted patients, including · Pneumonia (1) · Heart Attack (3) · Hip/Knee Replacement (5) · Heart Failure (6)
Numerical	Expected Rate	The expected 30-day readmission rate for that condition, based on the average performance of hospitals with similar patient populations.
	Predicted Rate	The hospital's predicted 30-day readmission rate for a specific condition, considering the hospital's patient mix and other factors.
	ERR	Predicted Rate / Expected Rate
	Payment	Average Payments for received healthcare
	Lower Estimate	Lower bound for estimated Payment
	Higher Estimate	Upper bound for estimated Payment

Table 2. Cluster Predictions Summary

	prediction	Avg_Excess_Readmission_Ratio	Avg_Payment	Avg_Readmission_State_indexed	Avg_Payment_State_indexed
1	1	1.0060836205955694	17134.409728718427	7.889850327408793	7.889850327408793
2	3	0.9859244240837679	17454.274345549737	30.26282722513089	30.26282722513089
3	2	0	840.427807486631	11.493048128342245	11.493048128342245
4	0	1.0011764375798236	17149.0155651341	7.271551724137931	7.271551724137931

Table 3. Cluster Predictions - Common State and Health Condition

prediction	mode_Payment_State_indexed	mode_Payment_Measure_Name_indexed
1	6.0	2.0
3	29.0	1.0
2	9.0	2.0
0	6.0	1.0